

Torch

```
torch.index_select()
torch.tensor.detach().cpu().numpy()
```

- `device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')`

gradient clipping simple solution to gradient explosion, works well practically

- `torch.nn.utils.clipgrad_norm(parameters, max_norm, norm_type=2)`
 - example: `torch.nn.utils.clipgrad_norm(retinanet.parameters(), 0.1)`

Tensor shapes

From Simple BEV repo: We maintain consistent axis ordering across all tensors. In general, the ordering is `B, S, C, Z, Y, X`, where

- B: batch
- S: Sequence (for temporal or multiview data)
- C: channels
- Z: depth
- Y: height
- X: width

The ordering stands even if a tensor is missing some dims, for example, plain images are `B, C, Y, X` (as is the pytorch standard).

Axis directions

- Z: forward
- Y: down
- X: right

This means the top-left of an image is "0.0", and coordinates increase as you travel right and down. `z` increases forward because it's the depth axis.

1. Chapter 2

```
1. import torch
   device = torch.device("cuda:0" if torch.cuda.is_available() else "cpu")

   net.to(device)
   images = images.to(device)
   labels = labels.to(device)
   output = net(images)
   loss = criterion(output, labels)
```

Pytorch

Tensor

1. From an interface perspective, operations on Tensors can be divided into two categories:
 1. `torch.function`, such as `torch.save`, etc.
 2. `tensor.function`, such as `tensor.view`, etc.
2. For ease of use, most operations on Tensors support both types of interfaces simultaneously, and this book does not make a specific distinction, for example, `torch.sum(a, b)` and `a.sum(b)` are functionally equivalent.
3. From a storage perspective, operations on Tensors can be divided into two categories:
 1. A function that does not modify its own data, such as `a.add(b)` will return a new Tensor as the result of the addition.
 2. A function that modifies its own data, such as `a.add_(b)` will still store the result if the addition in `a`, but `a` will be modified.

- Functions whose names end with an underscore are inplace functions, meaning they modify the caller's own data. This distinction is important in practical applications.

Creating Tensors

- There are many ways to create tensors in Pytorch, as shown in Table 3-1.

Function	Functionality
<code>Tensor(*sizes)</code>	Basic constructor
<code>tensor(data,)</code>	Constructor similar to <code>np.array()</code>
<code>ones(*sizes)</code>	Tensor with all ones
<code>zeros(*sizes)</code>	Tensor with all zeros
<code>eye(*sizes)</code>	The diagonal is 1, and the rest are 0
<code>arange(s, e, step)</code>	From s to e, the step size is step
<code>linspace(s, e, steps)</code>	From s to e, divide evenly into steps
<code>rand/randn(*sizes)</code>	uniform/standard distribution
<code>normal(means, std)/uniform(from, to)</code>	normal distribution/ uniform distribution
<code>randperm(m)</code>	random arrangement
<code>tensor.new_*/torch.*_like</code>	create a Tensor of the same size, filled with * type (e.g. <code>tensor_new_ones</code> , is filled with all ones), and with the same <code>torch.dtype</code> and <code>torch.device</code>

- All these creation methods allow you to specify the data type (`dtype`) and the storage device (CPU/GPU) during creation.
- The `tensor` function can be used to create a new Tensor. It can accept a list and create a new tensor based on the data in the list. It can also create a tensor based on the data in the list. It can also create a tensor based on a specified data shape and can pass in other tensors. Several examples will be given below.
- It's important to note that when `t.Tensor(*sizes)` creates a Tensor, the system doesn't immediately allocate space. It only calculates whether there's enough remaining memory and allocates space immediately after tensor creation. In practice we recommend using `torch.tensor` instead of `torch.Tensor`. Let's compare the differences between the two.
- `torch.Tensor` is a python class, and by default it is `torch.FloatTensor()`. Running `torch.Tensor([2, 3])` will directly call the `__init__()` constructor of the `Tensor` class, generating a single-precision floating-point `Tensor(FloatTensor)`.
- `torch.tensor` is a python function with the prototype `torch.tensor(data, dtype=None, device=None, requires_grad=False)`. `data` supports types such as list, tuple, array, and scalar. `torch.tensor()` directly copies data from `data` and generates the corresponding Tensor based on the original data type.
- Since `torch.tensor()` can generate a corresponding Tensor based on the data type, we recommend using the `torch.tensor()` method to create a new Tensor in practical applications.

Types of Tensor

- Tensor types can be further divided into device type and data type (`dtype`). Device type is divided into CUDA and CPU types, and numeric types include bool, float, and int.
- Tensor data types are shown in Table 3-2, and each data type has CPU and GPU versions. Readers can obtain the device type of a tensor using `tensor.device` and the data type using `tensor.dtype`.
- The default data type of a Tensor is `FloatTensor`. Readers can modify the default Tensor type using `torch.set_default_tensor_type` (if the default type is a GPPU Tensor, then all operations will be performed on the GPU) or `torch.set_default_dtype` (only accepts float input types).
- Understanding the type of a Tensor is helpful for analyzing memory usage. For example, a `FloatTensor` of shape (1000, 1000, 1000) has $1000 \times 1000 \times 1000 = 10^9$ elements, each occupy $32\text{bit} \div 8 = 4\text{byte}$ of memory/GPU memory. `HalfTensor` is specifically designed for GPU versions. With the same number of elements, `HalfTensor` uses only half the GPU memory of a `FloatTensor`.
- Therefore, using `HalfTensor` can greatly alleviate the problem of insufficient GPU memory. It should be noted that `HalfTensor` has limited numerical values and precision, which may lead to data overflow issues.
- Tensor data type

Data Type	dtype	CPU tensor	GPU tensor
32-bit floating-point	torch.float32 or torch.float	torch.FloatTensor	torch.cuda.FloatTensor
64-bit floating-point	torch.float64 or torch.double	torch.DoubleTensor	torch.cuda.DoubleTensor
16-bit half-precision floating-point	torch.float16 or torch.half	torch.HalfTensor	torch.cuda.HalfTensor
8-bit unsigned integer	torch.uint8	torch.ByteTensor	torch.cuda.ByteTensor
8-bit signed integer	torch.int8	torch.CharTensor	torch.cuda.CharTensor
16-bit signed integer	torch.int16 or torch.short	torch.ShortTensor	torch.cuda.ShortTensor
32-bit signed integer	torch.int32 or torch.int	torch.IntTensor	torch.cuda.IntTensor
64-bit signed integer	torch.int64 or torch.long	torch.LongTensor	torch.cuda.LongTensor
Boolean type	torch.bool	torch.BoolTensor	torch.cuda.BoolTensor

7. Common methods for converting between different types of tensors are as follows:

1. The most common approach is `tensor.type(new_type)`, but there are also shortcut methods such as `tensor.float()`, `torch.long()`, and `tensor.half()`.
2. Conversion between CPU tensors and GPU tensors is achieved using `tensor.cuda()` and `tensor.cpu()`, and `tensor.to(device)` can also be used.
3. Creating tensors of the same type: `torch.*_like` and `torch.new_*`. These two methods are suitable for writing device compatible code:
 1. `torch.*_like(tensorA)` generates a new tensor with the same attributes as `tensorA` (such as type, shape and CPU/GPU)
 2. `tensor.new_*(new_shape)` creates a new tensor with a different shape but the same attributes.

References:

1. Pytorch book
2. Modern Computer Vision with PyTorch: Explore deep learning concepts and implement over 50 real-world image applications, Kishore Ayyadevara.